7. Design a scene that allows the user to place a character on a table and interact with it in Unity.

PART 1: Create the Reading Table

1. Table Top

- Right-click → `3D Object → Cube`
- Rename → `TableTop`
- Scale:
    - X = 2
    - Y = 0.1
    - Z = 1
- Position:
    - Y = 1 (height above ground)

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2. Table Legs (4 legs)

- Right-click → `3D Object → Cube`
- Rename → `Leg1`
- Scale:
    - X = 0.1, Y = 1, Z = 0.1
- Duplicate (Ctrl + D) → make 4 legs
- Position them at corners:
    - Leg1 → (-0.9, 0.5, -0.4)
    - Leg2 → (0.9, 0.5, -0.4)
    - Leg3 → (-0.9, 0.5, 0.4)
    - Leg4 → (0.9, 0.5, 0.4)

OPTION 1 (BEST for marks): Build a Simple Human Character (using shapes)

Create Parts

1. Head

- `3D Object → Sphere`
- Rename → `Head`
- Scale:
    - X = 0.3, Y = 0.3, Z = 0.3

- Position:
    - X = 0, Y = **1.9**, Z = 0

---

2. Body

- `3D Object → Cube`
- Rename → `Body`
- Scale:
    - X = 0.5, Y = 0.8, Z = 0.3
- Position:
    - X = 0, Y = **1.4**, Z = 0

---

3. Legs (2)

- Cube → `LeftLeg`
    - Scale: (0.2, 0.7, 0.2)
    - Position: (-0.15, **0.8**, 0)
- Duplicate → `RightLeg`
    - Position: (0.15, **0.8**, 0)

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4. Arms (2)

- Cube → `LeftArm`
    - Scale: (0.2, 0.6, 0.2)
    - Position: (-0.45, **1.4**, 0)
- Duplicate → `RightArm`
    - Position: (0.45, **1.4**, 0)

□ Group Them

1. Create Empty Object → `Character`
2. Drag all parts inside it

Final Character Position (on table)

Set **Character (parent)**:

- X = 0
- Y = **1.05**
- Z = 0

☞ Now whole body sits correctly on table

Adjust the character so that it will be on the top of the table

If above character didn't work use following coordinates

STEP 1: Reset Parent (Character)

Select **Character (parent)** and set:

- **Position**
  - X = 0
  - Y = **1.05**
  - Z = 0
- **Scale**
  - X = 1, Y = 1, Z = 1

🎯 STEP 2: Set CHILD PARTS (LOCAL positions)

Now select each part **inside Character** and set these:

👤 Head

- Position:
  - X = 0
  - Y = **0.9**
  - Z = 0

▯ Body

- Position:
  - X = 0
  - Y = **0.4**
  - Z = 0

▯ Left Leg

- Position:

- X = -0.15
- Y = **0.0**
- Z = 0

Right Leg

- Position:
    - X = 0.15
    - Y = **0.0**
    - Z = 0

💪 Left Arm

- Position:
    - X = -0.35
    - Y = **0.4**
    - Z = 0

💪 Right Arm

- Position:
    - X = 0.35
    - Y = **0.4**
    - Z = 0

Add Movement to Character

1. Create Script

- Right-click in **Assets**
- Create → C# Script → name: `CharacterMove`

```
using UnityEngine;

public class CharacterMove : MonoBehaviour
{
    public float speed = 3f;

    void Update()
    {
        float h = Input.GetAxis("Horizontal"); // A/D or Arrow keys
        float v = Input.GetAxis("Vertical");   // W/S or Arrow keys

        Vector3 move = new Vector3(h, 0, v);
        transform.Translate(move * speed * Time.deltaTime);
    }
}
```

Attach Script

- Select **Character (parent object)**
- Click **Add Component**
- Select `CharacterMove`

☐ PART 6: Add Physics (Important)

1. Add Rigidbody

- Select **Character**
- Add Component → **Rigidbody**

Set:

- Use Gravity ✓
- Freeze Rotation X ✓
- Freeze Rotation Z ✓

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2. Prevent Falling

- In Rigidbody:
  ☞ Freeze Position Y ✓

PART 7: Place Character on Click

1. Create Script → `PlaceCharacter`

```
using UnityEngine;

public class PlaceCharacter : MonoBehaviour
{
    public float speed = 3f;

    void Update()
    {
        float h = Input.GetAxis("Horizontal");
        float v = Input.GetAxis("Vertical");

        Vector3 move = new Vector3(h, 0, v);
        transform.Translate(move * speed * Time.deltaTime, Space.World);
    }
}
```

2. Attach Script

- Create Empty Object → rename **GameManager**
- Add Component → `PlaceCharacter`
- Drag **Character → into script field**

PART 8: Table Collider (MANDATORY)

- Select **TableTop**
- Ensure:
  ✔ Box Collider is present

☞ Without this, clicking won't work

🎥 PART 9: Camera Setup

- Select **Main Camera**
- Adjust:
  - Position: (0, 2, -4)
  - Rotation: (20, 0, 0)

☞ Or press:
✔ **Ctrl + Shift + F**

▶ PART 10: Test Everything

1. Click **Play**
2. Click on table
   ☞ Character moves to that point
3. Use:

- **W/A/S/D or Arrow keys**
  ☞ Character moves

Switch Input System

1. Go to **Edit → Project Settings**
2. Click **Player**
3. Find **Active Input Handling**

4. Select:
   ☞ **Both**
5. Click **Apply**
6. Restart Unity

Keep gravity on in rigidbody of character

If want

5. Improve Ground Interaction

Select **Ground / Floor**:

- Ensure **Box Collider** exists

Optional:

- Create **Physics Material**
  - Friction = 0.6
  - Bounciness = 0

☞ Assign to ground → prevents sliding